CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 17: Circles

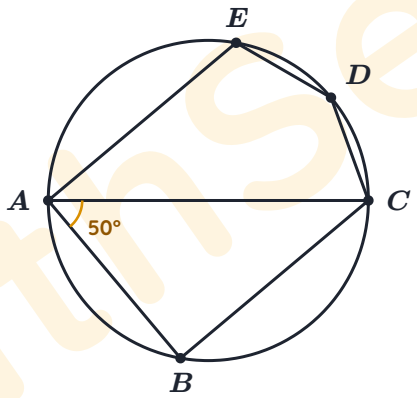
Previous year board questions, topic-wise: 7 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

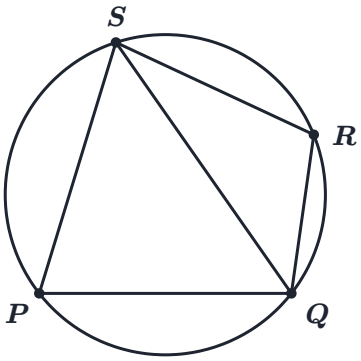
1 Cyclic quadrilaterals

1. A, E, C and B lie in order on a circle. Chords AC and BC are equal, BC is produced to D, and $\angle ACD = 120^\circ$. Find $\angle AEC$. (a) 30° (b) 60° (c) 90° (d) 120° ICSE 2025, 1 mark

2. $ABCDE$ is a pentagon inscribed in a circle. AC is a diameter, $BC \parallel AE$ and $\angle BAC = 50^\circ$. Find (a) $\angle ACB$ (b) $\angle EDC$ (c) $\angle BEC$. Then prove that BE is also a diameter. ICSE 2019, 4 marks

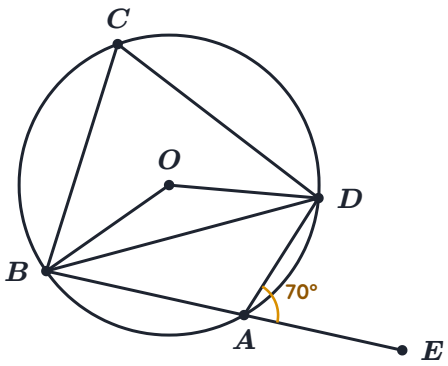


3. $PQRS$ is a cyclic quadrilateral with $\angle QPS = 73^\circ$, $\angle PQS = 55^\circ$ and $\angle PSR = 82^\circ$. Find (a) $\angle QRS$ (b) $\angle RQS$ (c) $\angle PRQ$. ICSE 2018, 4 marks



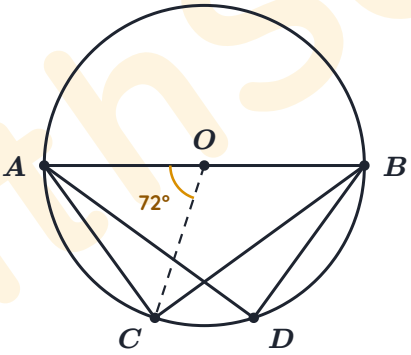
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4. $ABCD$ is a cyclic quadrilateral in a circle with centre O. BA is produced to E and $\angle DAE = 70°$. Find (a) $\angle BCD$ (b) $\angle BOD$ (c) $\angle OBD$. ICSE 2017, 4 marks



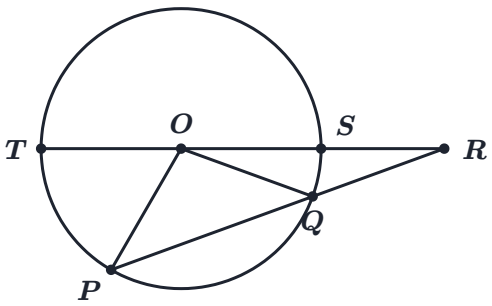
2 Angle at the centre and at the circumference

5. AB is a diameter of a circle with centre O. C and D are points on the circle on the same side of AB, with $AC = BD$ and $\angle AOC = 72°$. Find (a) $\angle ABC$ (b) $\angle BAD$ (c) $\angle ABD$. ICSE 2020, 3 marks

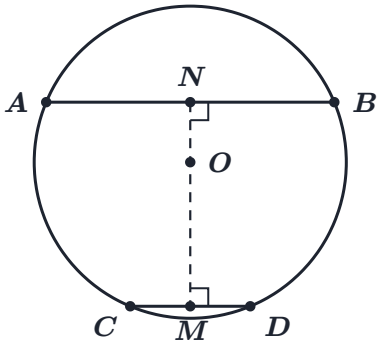


3 Arcs, chords and angles

6. O is the centre of a circle. A line through O meets the circle at T and S and is produced to R. A line from R cuts the circle at Q and P, with Q nearer to R. $QR = OP$ and $\angle ORP = 20°$. Find $x = \angle TOP$. ICSE 2018, 3 marks



7. AB and CD are parallel chords on opposite sides of the centre of a circle of radius 13 cm. $AB = 24$ cm and $CD = 10$ cm. Find the distance between the chords. ICSE 2017, 3 marks



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